

BARYON ASYMMETRY AS CURVATURE-RATIO LOCK-IN: RATIO-LEVEL FIRST PRINCIPLES IN THE HQIV LEAN LIBRARY

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ABSTRACT. Position in the publication chain. This is the sixth paper in the HQIV preprint series and the third of the Tier-2 Standard-Model layer. Tier 0 fixes the framework [6, 8, 10]; Tier 1 supplies the variational and lattice spine [11, 9, 7]; Tier 2 adds electroweak and mass closure [5, 4]. The present note states the strongest *ratio-level* baryogenesis theorem currently supported by the library: baryon-asymmetry bookkeeping is not an independent knob—it is the same partial curvature imprint as the growth channel, triality CP-bias scaffolds, and the HQVM Friedmann compatibility package.

Result. The formal spine is entirely ratio-level: discrete shell combinatorics define a cumulative curvature integral $I(n)$; horizon ratios $\Omega_k(n; N) = I(n)/I(N)$ are machine-checked with proved monotonicity in n at fixed N ; the lock-in shell $m_{\text{lockin}} = M = \text{referenceM}$ calibrates $\Omega_k = 1$ at the lock-in horizon; and the baryon-asymmetry readout $\eta(n; N)$ is *defined* to share that same normalisation, pointwise in (n, N) —the input convention the companion BBN note [3] takes at lock-in. Light-element abundances are *not* claimed here; that companion documents the BBN Lean stack (network readout, epoch ladder, integrated witness). Reader-friendly anchors with clickable Lean links are in Appendix A.

Single absolute anchor (quarantined). The observed-order scale η_{paper} appears in exactly one module (the [baryogenesis \$\eta\$ paper module](#)) and is wired only through the witness layer. Ratio laws and lock-in calibration are proved without it. Under the proton lock-in witness discipline of [5, 7], laboratory η is a comparison target, not a simultaneous fit row alongside other PDG inputs.

Patch-ontology reader contract. Per the patch-theory reader contract below, the discrete patch layer is the ontology. Smooth-manifold transport, washout, and sphaleron yield are comparison-layer physics targets—not foundational requirements and not a continuum-completion programme.

Dynamic bulk witnesses (v2 addition). A per-shell matter integrator now derives comparison-layer Ω_b and η without importing η_{paper} as an input. Central readout: $\eta_{10} = 5.60$ from `data/dynamic_bulk_bbn_v2.json` (witness `scripts/hqiv_dynamic_bulk_bbn.py`). Different legitimate stages bracket $\eta_{10} \approx 5.6\text{--}6.4$: legacy lock-in anchor (6.10), early dynamic-baryogenesis correction (6.28), and comparison-layer H_0 propagation at fixed Ω_b (6.04 at $H_0 = 70$, 6.39 at $H_0 = 72 \text{ km s}^{-1} \text{ Mpc}^{-1}$). Casimir inside/outside balance sets the temperature-relative vev; bound-state binding release modulates BBN Q values; the BBN network advances by a shell reaction-opportunity factor, not by Hubble rate as an input clock.

Status. Proved in this note: monotonicity of I and Ω_k , lock-in calibration, parallel η laws, triality tilt averaging, and GR/O–Maxwell compatibility. Computed witnesses (not yet full Lean certificates): dynamic bulk η band and shell-opportunity BBN integration (§6). Companion [3]: BBN umbrella and network/epoch/witness modules consuming lock-in η . Not claimed here: derivation of η_{paper} from associator phases, SM chemical potentials, or a unique closed-form η without comparison-layer inputs.

Patch QFT and discrete numerics (reader contract). HQIV (Horizon-Quantized Informational Vacuum) is formulated as a *patch theory*: quantum fields, actions, and physical readouts are attached to discrete patches—shell indices $m \in \mathbb{N}$, finite spacetime index types (e.g. $\text{Fin } 4$), and horizon-limited charts—rather than to a hypothetical fundamental smooth spacetime obtained as a mesh-refinement or ultraviolet limit. Accordingly, when this text refers to “QFT,” it means *patch QFT*: patching rules, finite measurement ledgers, and variational identities on the discrete spine; it is not the claim that the theory is *defined* by recovering ordinary continuum quantum field theory through such a limit. The numbers that admit the sharpest statements—mass and coupling

ladders, curvature ratios, shell thermodynamics, and rational witnesses in `HQIV_LEAN`—are objects of *discrete mathematics* on $\mathbb{N}$ (combinatorial growth, cumulative simplex ledgers) together with the ordered field $\mathbb{R}$ as used in the proof assistant for analysis, bounds, and phenomenological display. Smooth-manifold or continuum field notation, when it appears, is a *translation layer* for comparison with standard physics literature; it does not supersede the discrete definitions.

A continuum is not required, and in places would destroy these results. The discrete patch layer is *not* a numerical approximation to a more fundamental continuum theory awaiting future construction; it is the ontology. Where the literature treats “the continuum limit” as the goal, HQIV treats it as a *calculation approximation* for comparison with standard formulas. The continuum form is an optional convenience for comparison; the proved discrete statement is what carries the physics. In several load-bearing places—the curvature imprint $\delta_E(m)$ at shell m , the harmonic ladder masses $m_\tau \rightarrow m_\mu \rightarrow m_e$, the lock-in vev $v = \sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$, the proton anchor at $m = \text{referenceM} = 4$, the baryon-asymmetry $\eta \propto 6^7 \sqrt{3}/(m+1)$, and the rational coefficients $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$ —taking a literal $m \rightarrow \infty$ or sub-Planck continuum limit would *erase* the discrete index that is doing the predictive work. Continuum forms of these objects are therefore not preferred refinements; they are degenerate, information-losing readouts of the same patch data.

An exception: $\mathfrak{so}(8)$ as a de-facto continuum algebra from a discrete construction. Principled exception to the discrete-only stance: the closed Lie algebra $\mathfrak{so}(8) \supseteq G_2 \cup \{\Delta\}$ generated from the Fano octonion left-multiplication table and the phase-lift generator $\Delta \in (e_1, e_7)$. Although $\mathfrak{so}(8)$ is a continuous (28-dimensional real) Lie algebra, it is built here entirely from *discrete* ingredients (the seven imaginary octonion units, a finite Fano table, and one $\pi/2$ rotation), and its closure to 28 dimensions is a finite linear-algebra certificate (see the `Hqiv.S08Closure` / `Hqiv.S08ClosureInterface` stack). The Lie-group structure is therefore a *de-facto continuum algebra obtained from a discrete construction*: continuous in parameter space, but with no continuum spacetime, no continuum measure, and no UV limit attached. When this manuscript or its companions write $\exp(t\Delta) \in \text{SO}(8)$ or treat $\mathfrak{so}(8)$ rotations as smooth, those are statements internal to a finite-dimensional Lie group whose generators are certified by discrete data; they are *not* a continuum-spacetime commitment and do not require a continuum-QFT scaffolding.

1. SCOPE AND CLAIM DISCIPLINE

The HQIV programme ties discrete null-shell counting, a curvature imprint exponent $\alpha = 3/5$ (lattice-coherent with Brodie’s horizon thermodynamics [1]), and an eight-dimensional octonionic carrier with machine-checked $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$ data [10, 11]. The octonionic-action paper [9] documents how the same (α, ϕ) slots feed discrete O–Maxwell stationarity and an HQVM Friedmann *compatibility* identity; the SM Lagrangian bridge [4] packages the same ledger at sector level.

This note answers: **what baryogenesis statement is already a theorem, and what is still interpretive physics?** Geometry and ratios live in the [baryogenesis core module](#) and the [octonionic lightcone module](#); the η_{paper} anchor is isolated in the [baryogenesis \$\eta\$ paper module](#) and wired only through the [baryogenesis witness module](#).

- (1) **Proved in Lean (formal spine).** Shell indices, ladder temperatures, combinatorial curvature norm $\mathcal{N}_{\text{curv}}$, per-shell imprint $\delta_E(m)$, cumulative integral $I(n)$ with strict monotonicity (the [curvature-integral strict-monotonicity theorem](#)), horizon ratios $\Omega_k(n; N)$ with monotonicity, strict monotonicity, and the bound $\Omega_k \leq 1$ for $n \leq N$ (the [horizon-ratio monotonicity pack](#)), lock-in calibration $\Omega_k(M; M) = 1$ (the [lock-in \$\Omega_k\$ calibration](#)), parallel $\eta(n; N)$ with matching monotonicity lemmas (the [\$\eta\$ monotonicity pack](#)), and the identity $\eta = \eta_{\text{paper}} \cdot \Omega_k$ at fixed horizon (the [shared-normalisation theorem](#)).

- (2) **Proved algebraic packaging.** Triality-sensitive increments from the [rapidity CP-bias](#) in the [trialty-rapidity-well module](#); affine tilt averaging (the [trialty tilt average identity](#)); three-representation averaging back to the rapidity well (the [trialty increment average identity](#)).
- (3) **Structural compatibility (not transport).** The [GR-from-Maxwell module](#): shared ϕ , shared α , and homogeneous HQVM Friedmann equivalence—a compatibility bridge, not a washout-limited yield theorem.
- (4) **Companion: BBN [3].** The BBN Lean stack—umbrella certificate, network-from-weights readout, epoch ladder, and integrated witness (Appendix A, [BBN programme modules](#))—is documented in the companion note; it consumes lock-in η from this note and is *not* part of this note’s proof graph. Coc et al. semi-analytic fits in the library are labelled comparison-only, not HQIV inputs.
- (5) **Yield-level SM physics (not Lean).** Associator/vorticity sources, QCD-window maximisation, and sphaleron rates remain narrative targets beyond both this note and [3]. No continuum-spacetime limit is invoked or required.

2. DISCRETE SHELLS, TEMPERATURES, AND THE BARYOGENESIS WINDOW

Fix the HQIV discrete shell index $m \in \mathbb{N}$ as in [10, 11]. The library defines a QCD onset shell index m_{QCD} and a lock-in shell

$$m_{\text{lockin}} := m_{\text{QCD}} + \text{stepsFromQCDToLockin},$$

aligned with $M = \text{referenceM}$ (the same lock-in convention used throughout the formalisation and pinned by the proton lock-in witness in [5, 7]). Temperatures on the ladder are $T(m) = 1/(m + 1)$ in natural units (the [QCD and lock-in temperature closures](#) in the baryogenesis core module).

The finite shell set used for discrete bookkeeping is the explicit [baryogenesis shell window](#) (a `Finset` interval from m_{QCD} through a few shells after lock-in). This is the honest discrete substitute for a continuum “GeV window”: it is *not* proved from washout maximisation in Lean; it is the packaged combinatorial stage on which curvature ratios are evaluated.

3. CURVATURE IMPRINT, $\mathcal{N}_{\text{curv}}$, AND δ_E

Let $\sigma(m)$ denote the shell-shape factor (the [shell-shape factor](#)), built from the same α -weighted logarithmic curvature density used in the growth trilogy. The combinatorial curvature norm is the exact real scalar $\mathcal{N}_{\text{curv}} = 6^7 \sqrt{3}$, proved in the lightcone module from the structural “six cube directions, seven imaginary octonion directions, half-diagonal $\sqrt{3}$ ” bookkeeping (the [curvature-norm structural theorem](#) and related lemmas). The per-shell imprint is $\delta_E(m) = \mathcal{N}_{\text{curv}} \cdot \sigma(m)$ (the [QCD-shell imprint](#) specialises at m_{QCD}).

Structural rigidity (octonionic vs. quaternionic comparison). Lean also proves that a natural quaternionic Maxwell comparison norm undershoots the HQIV octonionic norm by a rigid factor 1296 shellwise (the [octonion-quaternion imprint ratio theorem](#)). This supports the narrative that $\mathcal{N}_{\text{curv}}$ is not an arbitrary normalisation: it is the imprint constant selected by the octonionic lift inside the same eight-dimensional carrier used for gauge closure.

4. HORIZON RATIOS Ω_k AND LOCK-IN CALIBRATION

Define the cumulative partial integral

$$I(n) := \sum_{m=0}^{n-1} |\sigma(m)|$$

(the [curvature integral](#)). For horizons N with $I(N) > 0$, the library defines $\Omega_k(n; N) := I(n)/I(N)$. At the lock-in shell, Lean proves positivity $I(m_{\text{lockin}}) > 0$ and the calibration $\Omega_k(m_{\text{lockin}}; m_{\text{lockin}}) = 1$ (the [lock-in \$\Omega_k\$ calibration](#)). This is the precise sense in which “spatial curvature lock-in” is a

theorem: normalisation of the discrete imprint integral at the declared reference horizon, not an experimental input.

Proposition 4.1 (Monotonicity and approach to lock-in, Lean). *Assume $I(N) > 0$. Then for all $n_1 \leq n_2$, $\Omega_k(n_1; N) \leq \Omega_k(n_2; N)$, with strict inequality when $n_1 < n_2$ (the [horizon-ratio monotonicity pack](#)). If additionally $n \leq N$, then $\Omega_k(n; N) \leq 1$. In particular, along shells $n \leq m_{\text{lockin}}$ with horizon $N = m_{\text{lockin}}$, the ratio $\Omega_k(n; m_{\text{lockin}})$ increases strictly toward 1 as n approaches m_{lockin} from below.*

5. BARYON ASYMMETRY η AS THE SAME NORMALIZATION (OPTIONAL ABSOLUTE ANCHOR)

Without importing any experimental η . Everything in the baryogenesis core module is logically independent of PDG-scale baryon asymmetry.

With a single absolute anchor (quarantined). The constant η_{paper} appears *only* in the [baryogenesis \$\eta\$ paper module](#). The witness module defines $\eta(n; N) := \eta_{\text{paper}} \cdot I(n)/I(N)$ when $I(N) > 0$ (and a conservative fallback otherwise), and proves the parallel identities to the Ω_k ratio (the [horizon \$\eta\$ readout](#) and [self-horizon \$\eta\$ identity](#) at $n = N$).

The conceptual punchline is already a theorem:

$$\eta(n; N) = \eta_{\text{paper}} \cdot \Omega_k(n; N)$$

(the [shared-normalisation theorem](#)). At fixed horizon, baryogenesis is not a second independent imprint law: it is the same partial-integral ratio as Ω_k , scaled by one overall anchor.

Same monotonicity and bounds as Ω_k . Lean also proves $\eta(n_1; N) \leq \eta(n_2; N)$ for $n_1 \leq n_2$ (the [\$\eta\$ monotonicity pack](#)), and $\eta(n; N) \leq \eta_{\text{paper}}$ whenever $n \leq N$. Thus η_{paper} is both the absolute scale at lock-in and an upper envelope for readouts inside the horizon—a direct consequence of shared monotonicity with Ω_k .

Honesty boundary (first principles vs. calibration). Lean does not currently derive η_{paper} from associator phases or SM chemical potentials. What *is* first-principles inside the formalisation is the shared ratio law and the lock-in calibration of Ω_k . Attaching the observed η scale is a *single-point calibration* quarantined to one module—analogueous to how the mass and Kirchhoff papers treat amplitude normalisation while keeping imprint *shape* combinatorially rigid.

6. DYNAMIC BULK MATTER AND METHOD RANGE FOR η

The ratio spine of §1–§5 fixes how $\eta(n; N)$ is *normalized* once an absolute anchor is declared. A separate Python witness chain now asks what η is when that anchor is *not* imported: baryon budget is accumulated shell-by-shell from curvature imprint, Casimir inside/outside balance, and composite-trace binding feedback, then converted to η only at a late comparison layer using present-day H_0 , T_{CMB} , and Ω_b .

Per-shell integrator (v3). The calculator `scripts/hqiv_dynamic_bulk_bbn.py` evolves shells $m_{\text{QCD}} \leq m \leq m_{\text{lock}}$ with emergent hadronic lock-in at $m = 4$ ($\xi_{\text{lock}} = 5$). At each step it reads shell shape $\sigma(m)$, inner/outer curvature proxies, the continuous-chart curvature ratio $\Omega_k(\xi)$ normalised to lock-in, a temperature-relative vev factor from T12/T13 Casimir balance, and dynamic proton mass and nuclear Q values from the 8×8 composite-trace network. Matter increments

$$\Delta S_m = \sigma(m) f_{\text{vev}}(\xi_m) + w_{\text{strong}} f_{\text{bind}}(m, \xi_m) \Omega_k(\xi_m)$$

accumulate into a cumulative source $C = \sum_m \Delta S_m$. Baryon and total matter fractions follow

$$\Omega_m = \gamma C / (C + \rho_{\text{rad}}), \quad \Omega_b = \Omega_m \cdot w_{\text{strong}} \cdot f_{\text{color}},$$

with HQIV structural weights $\gamma = 2/5$, $w_{\text{strong}} = 4/8$, and a colour-singlet projector $f_{\text{color}} = 1/3$ (trace-derived sweep below). The witness reports $C = 2.085$, $\Omega_b = 0.0451$, $\Omega_m = 0.270$ at lock-in.

Comparison-layer η . Only after the shell ladder is integrated does the witness form

$$\eta = \frac{n_b}{n_\gamma} = \frac{\Omega_b \rho_{\text{crit}}}{m_p n_\gamma(T_{\text{CMB}})},$$

with default $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $T_{\text{CMB}} = 2.7255 \text{ K}$. This is **not** part of the proved Lean ratio graph; it is an auditable late-universe comparison readout. Lean modules [Dynamic BBN and baryogenesis](#) and [shell reaction opportunity](#) now mirror the binding-release and shell-clock factors used in the witness.

Why η_{10} moved: $6.09 \rightarrow 6.36 \rightarrow 5.59$. The three numbers are not random drift; they label different stages of the programme:

- **6.10 (legacy anchor)**. η_{paper} in the quarantined [baryogenesis \$\eta\$ paper module](#); still the lock-in calibration used by the fixed- η BBN witness.
- ≈ 6.28 – 6.36 (**intermediate dynamic paths**). Early `hqiv_dynamic_bbn_baryogenesis.py` runs with `eta_at_horizon_dynamic` and binding-curvature correction ($\eta_{10} = 6.28$); alternatively, holding the dynamic bulk Ω_b fixed and raising only the comparison H_0 to 70 – $72 \text{ km s}^{-1} \text{ Mpc}^{-1}$ gives $\eta_{10} = 6.04$ – 6.39 .
- **5.60 (current bulk v3 central value)**. Full per-shell Ω_b accumulation with dynamic vev and binding feedback, then comparison-layer conversion at $H_0 = 67.4$.

None of these replaces the proved identity $\eta(n; N) = \eta_{\text{paper}} \cdot \Omega_k(n; N)$; they test how much of the observed scale can be *derived* before the single-point anchor is applied.

TABLE 1. Comparison-layer $\eta_{10} \equiv \eta \times 10^{10}$ from different HQIV stages and inputs. Errors on the dynamic-bulk central value propagate $H_0 = 67.4 \pm 2.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ at fixed Ω_b ; other rows are deterministic readouts at stated inputs. Observed band from BBN+*CMB* synthesis [2].

Method	η_{10}	uncertainty	Notes
Observed (BBN+ <i>CMB</i>)	6.10	± 0.06	Comparison target [2]
Lean lock-in anchor (η_{paper})	6.10	—	Quarantined calibration; not derived here
Early dynamic baryogenesis	6.28	—	<code>eta_at_horizon_dynamic</code> + binding correction
Dynamic bulk v3 (primary)	5.60	± 0.42	Per-shell Ω_b ; H_0 error propagation
$H_0 = 70$ at fixed Ω_b	6.04	—	Comparison-layer H_0 sensitivity
$H_0 = 72$ at fixed Ω_b	6.39	—	Upper bracket of H_0 sweep
$\rho_{\text{rad}} = 0.5$ vs. 1.0	6.68 / 5.60	—	Radiation-floor sensitivity on shell source

Casimir-modulated binding and BBN clock. Bound-state Casimir trapping in the early curvature background changes effective nuclear Q values during BBN. The witness uses a curvature-temperature binding-release factor

$$f_{\text{bind}}(T) = \exp\left(-\gamma w_{\text{strong}} \frac{s}{1+s}\right), \quad s = \frac{\Omega_k(\xi) - \Omega_k(\xi_{\text{lock}})}{\log(\xi/\xi_{\text{lock}})},$$

applied to lock-in composite-trace Q 's rather than raw $\xi = T_{\text{Pl}}/T$ vev blow-up. For BBN network integration the native clock is a **shell reaction opportunity**

$$\mathcal{O}_{\text{shell}} = \Delta \log \xi \cdot [\log(\xi/\xi_{\text{lock}})]^3 \Omega_k(\xi)^{\gamma w_{\text{strong}}},$$

optionally perturbed by binding-sourced curvature feedback. Standard radiation-dominated $H(T)$ is retained only as a diagnostic comparison to textbook BBN language (`legacy_H_cooling` mode).

TABLE 2. BBN abundance sensitivity at the dynamic-bulk $\eta = 5.60 \times 10^{-10}$: network driver and Q model. Abundance certificates remain in [3]; rows here document the method spread tied to this baryogenesis witness update.

Network driver	Y_p	D/H	Notes
Shell opportunity + dynamic Q	0.264	2.23×10^{-5}	Primary v3 path
Legacy $H(T)$ + dynamic Q	0.265	2.33×10^{-5}	Comparison to standard cooling
Legacy $H(T)$ + fixed lock-in Q	0.249	5.3×10^{-10}	Fixed- η_{paper} baseline
Observed band	0.244 ± 0.004	$(2.53 \pm 0.04) \times 10^{-5}$	[2]

Honest error budget. The dynamic bulk central value sits $\approx 8\%$ below the observed η_{10} at default H_0 . Propagating only the comparison-layer H_0 uncertainty ($\pm 2.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$) gives $\eta_{10} = 5.60 \pm 0.42$. A parallel early-dynamic baryogenesis correction and an H_0 -only raise bracket 6.0–6.4 without changing the per-shell source. Remaining upward levers visible in the witness—not yet promoted to theorems—include a trace-derived colour-singlet projector above 1/3 and the Lean `baryogenesis_binding_curvature_correction` term in `eta_at_horizon_dynamic`. Downward levers include a larger effective radiation floor in the shell partition and stopping the integrator before additional post-lock-in shell accumulation.

Witness discipline. Machine-checked JSON: `data/dynamic_bulk_bbn_v2.json` (bulk + BBN v3); `data/bbn_witnesses_dynamic.json` (early dynamic demo). Recompute with `python3 scripts/hqiv_dynamic_bulk_bbn.py`.

7. CP BIAS WITHOUT NAMING η : TRIALITY COUPLING

The `triality-rapidity-well` module defines

$$\text{rapidityCPBias}(m) := \Omega_k(m; m_{\text{lockin}}) - 1,$$

the `rapidity CP-bias`—a pure curvature-ratio deviation at the lock-in reference horizon. It then builds representation-sensitive triality increments by multiplying baseline barrier surfaces by $(1 + \text{rapidityCPBias}(m) \cdot w_{\text{rep}})$ with explicit $w_{\text{rep}} \in \{0, +1, -1\}$ on the three Spin(8) triality channels, and proves exact three-rep averaging back to the rapidity well.

Auditable zero-sum tilt. Writing the weights as $w_{8_v} = 0$, $w_{8_{s+}} = +1$, $w_{8_{s-}} = -1$, Lean proves the affine factors average identically to unity:

$$(1) \quad \frac{1}{3} \sum_{\text{rep} \in \{8_v, 8_{s+}, 8_{s-}\}} (1 + w_{\text{rep}} \cdot \text{rapidityCPBias}(m)) = 1$$

(the `triality tilt average identity`). The full triality increments average to the rapidity-written well surface (the `triality increment average identity`).

This is the cleanest Lean-backed “CP-odd bias is tied to the same curvature-ratio channel” statement that does *not* import η_{paper} . It is still **not** a replacement for CPT/sphaleron/washout dynamics; it is a structured, finite-dimensional coupling scaffold consistent with the lock-in ratios.

8. LINK TO HQVM FRIEDMANN AND O-MAXWELL (COMPATIBILITY, NOT YIELD)

The `GR-from-Maxwell` module packages that the same auxiliary ϕ appears in the HQVM lapse story and the O-Maxwell bookkeeping (the `shared- $\phi$  identity`), and that the same lattice $\alpha = 3/5$ enters $G_{\text{eff}}(\phi) = \phi^\alpha$ on the Friedmann side (the `shared- $\alpha$  identity`). The homogeneous Friedmann identity is packaged as an explicit algebraic equivalence (the `HQVM Friedmann compatibility theorem`).

This supports the *narrative* that cosmological background evolution used in companion numerics is not stitched from unrelated parameters: it is the same (α, ϕ) ledger as the discrete gauge sector [9]. It does *not*, by itself, integrate Boltzmann equations for baryon number violation in the SM.

9. WHAT WOULD COMPLETE A “FULL DYNAMICAL DERIVATION”

To go beyond ratio bookkeeping and triality scaffolds, a future formalisation would need explicit discrete transport theorems on the same shell indices, for example:

- a well-posed chemical-potential / Boltzmann layer tied to shell indices;
- a washout bound as a theorem about monotone functionals of $I(n)$ under a declared interaction kernel;
- a provable map from non-associative phase holonomy to a signed baryon production density, with finite totals controlled by $I(n)$.

None of these are discharged in the current repository. Smooth-manifold rewrites of such transport are comparison-layer targets only—not a continuum-completion programme. The present manuscript is the maximal honest baryogenesis paper aligned to actual Lean artifacts.

COMPANION: BIG-BANG NUCLEOSYNTHESIS [3]

This note stops where BBN begins at the *ratio-level* lock-in calibration $\Omega_k(M; M) = 1$ and the shared $\eta(n; N) = \eta_{\text{paper}} \cdot \Omega_k(n; N)$ readout. The companion BBN preprint documents the library stack: the [BBN umbrella module](#) re-exports a network readout from composite-trace binding and isotope-ladder weights ([BBN network-from-weights module](#)), an epoch ladder ([BBN epoch modules](#)), and an integrated witness certificate ([BBN witness module](#)) checked against `data/bbn_witnesses.json`. Standard semi-analytic Y_p and D/H reference fits (Coc et al.) appear only as *comparison layers*, not as HQIV-derived inputs.

The present v2 update adds the dynamic-bulk path that *feeds* BBN: per-shell Ω_b and η from `hqiv_dynamic_bulk_bbn.py`, Casimir-modulated binding Q ’s, and shell reaction opportunity in place of $H(T)$ as the integrator clock (Tables 1–2). Those witnesses are documented here because they determine the comparison-layer η passed into the BBN network; full abundance theorems remain in [3].

10. CONCLUSION

The first-principles content formalised today is:

- (1) a rigid combinatorial imprint norm $\mathcal{N}_{\text{curv}} = 6^7 \sqrt{3}$ with an octonionic/quaternionic contrast theorem;
- (2) a discrete cumulative curvature integral $I(n)$ and horizon ratios $\Omega_k(n; N)$ with proved strict monotonicity of I , monotonicity (and strict monotonicity) of Ω_k in the readout shell, the bound $\Omega_k(n; N) \leq 1$ for $n \leq N$, and lock-in calibration $\Omega_k = 1$ at m_{lockin} ;
- (3) a parallel $\eta(n; N)$ law sharing the *identical* normalization as $\Omega_k(n; N)$, with matching monotonicity lemmas and $\eta(n; N) \leq \eta_{\text{paper}}$ for $n \leq N$, and with η_{paper} quarantined to a single module;
- (4) a triality representation scaffold coupling curvature-ratio bias to CP-sensitive increments, without importing η_{paper} . This scaffold ties CP-odd increments to the same curvature-ratio channel used for Spin(8) closure and measurement; it does *not* derive the bias itself from non-associativity of the octonion product (that remains a Tier-IV target).
- (5) a GR/O–Maxwell compatibility package tying (α, ϕ) across sectors;
- (6) **(v2 witnesses)** a dynamic bulk integrator deriving comparison-layer $\eta_{10} \approx 5.6\text{--}6.4$ from per-shell curvature, Casimir vev, and binding feedback, with shell-opportunity BBN integration yielding $Y_p \approx 0.264$ and $D/H \approx 2.2 \times 10^{-5}$ at the dynamic η (Tables 1–2).

That package is already enough for a mathematically disciplined “baryogenesis stands on the same shoulders as Spin(8) closure” paper, now augmented by an explicit method-range audit for the observed η band. The natural next formal manuscript in the programme is the BBN stack above; the remaining distance to textbook electroweak baryogenesis beyond that is washout, sphaleron integration, and associator-derived CP violation—yield-level dynamics not yet in the Lean corpus. Closing the $\approx 8\%$ gap between the dynamic-bulk central η_{10} and the observed value is expected to come from trace-derived baryon projectors and binding-sourced baryogenesis corrections already scaffolded in Lean, not from refitting η_{paper} .

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AI-assisted drafting and repository tooling were used during manuscript preparation. All formal claims were checked against the cited Lean modules in the HQIV-LEAN library [12].

APPENDIX A. LEAN MODULE INDEX

The Lean development lives at <https://github.com/HQIV/hqiv-lean> [12]. Reader-friendly anchors in the body link here.

Octonionic lightcone module.:

Shell shape, curvature integral, Ω_k monotonicity pack, combinatorial curvature norm. [Hqiv/Geometry/OctonionicLightCone.lean](#).

Shell-shape factor.:

shell_shape.

Curvature integral.:

$I(n) = \sum_{m < n} |\sigma(m)|$. curvature_integral.

Curvature-integral strict monotonicity.:

curvature_integral_strict_mono.

Curvature-norm structural theorem.:

curvature_norm_determined_by_structure.

QCD-shell imprint.:

deltaE_at_QCD_shell.

Octonion–quaternion imprint ratio.:

deltaE_eq_1296_mul_deltaE_quaternionicCandidate.

Baryogenesis core module.:

Shell indices, temperatures, Ω_k without η . [Hqiv/Physics/BaryogenesisCore.lean](#).

QCD and lock-in temperature closures.:

T_QCD_closed, T_lockin_closed.

Baryogenesis shell window.:

baryogenesisShells.

Horizon-ratio monotonicity pack.:

omega_k_at_horizon_mono, omega_k_at_horizon_strict_mono, omega_k_at_horizon_le_one_of_le.

Lock-in Ω_k calibration.:

omega_k_lockin_calibration.

Baryogenesis η paper module.:

η_{paper} only. [Hqiv/Physics/BaryogenesisEtaPaper.lean](#).

Baryogenesis witness module.:

η monotonicity and shared normalisation. [Hqiv/Physics/BaryogenesisWitness.lean](#).

Horizon η readout.:

eta_at_horizon_eq.

Self-horizon η identity.:`eta_at_horizon_self.` **η monotonicity pack.:**`eta_at_horizon_mono, eta_at_horizon_strict_mono, eta_at_horizon_le_eta_paper.`**Shared-normalisation theorem.:**`baryogenesis_same_normalization_as_omega_k.`**Triality-rapidity-well module.:**`Hqiv/Physics/TrialityRapidityWellEquivalence.lean.`**Rapidity CP-bias.:**`rapidityCPBias.`**Triality tilt average identity.:**`triality_cp_tilt_factors_average_eq_one.`**Triality increment average identity.:**`cpSensitiveTrialityIncrement_threeRep_average_eq_rapidityWell.`**GR-from-Maxwell module.:**`Hqiv/Physics/GRFromMaxwell.lean.`**Shared- ϕ identity.:**`same_phi_in_0_Maxwell_and_HQVM.`**Shared- α identity.:**`same_alpha_in_0_Maxwell_and_HQVM.`**HQVM Friedmann compatibility.:**`0_Maxwell_compatible_with_HQVM_GR_homogeneous.`**BBN companion [3] (not in this note's graph).:**

Umbrella + network + epoch + witness; consumes lock-in η . See entries below.

BBN umbrella module.:

Combined vital readout certificate. `Hqiv/Physics/BigBangNucleosynthesis.lean`.

BBN network-from-weights module.:

Primary light-element readout from HQIV weights. `Hqiv/Physics/BBNNetworkFromWeights.lean`.

BBN epoch modules.:

Epoch ladder and evolution. `Hqiv/Physics/BBNEpochNetwork.lean`, `Hqiv/Physics/BBNEpochEvolution.lean`.

BBN witness module.:

Integrated witness + Python data/bbn-witnesses.json. `Hqiv/Physics/BBNWitness.lean`.

Dynamic BBN and baryogenesis module.:

Curvature-temperature binding release, effective BBN Q 's at temperature, shell reaction opportunity with binding feedback. `Hqiv/Physics/DynamicBBNBaryogenesis.lean`.

Shell reaction opportunity.:

`bbnShellReactionOpportunity, bbnCurvatureOpportunityFactor` in `Hqiv/Physics/BBNEpochNetwork.lean`.

Dynamic bulk witness (Python).:

`scripts/hqiv_dynamic_bulk_bbn.py` $\rightarrow$ `data/dynamic_bulk_bbn_v2.json`.

APPENDIX B. LEAN DEPENDENCY SKETCH

`OctonionicLightCone`

```
(shell_shape, curvature_integral, curvature_integral_strict_mono,
 omega_k_at_horizon_mono / _strict_mono / _le_one_of_le,
 curvature_norm_combinatorial, deltaE, referenceM, qcdShell, ...)
```

|
v

`BaryogenesisCore`

```

(m_QCD, m_lockin, T_QCD, T_lockin, baryogenesisShells,
 omega_k_lockin_calibration, baryogenesis_vital_omega_T_no_eta)
|
+-----> TrialityRapidityWellEquivalence
|         (rapidityCPBias from omega_k_at_horizon;
|         triality_cp_tilt_factors_average_eq_one;
|         cpSensitiveTrialityIncrement_threeRep_average_eq_rapidityWell)
|
+--> BaryogenesisEtaPaper (eta_paper only)
|
v
BaryogenesisWitness
(eta_at_horizon_mono, eta_at_horizon_strict_mono,
 eta_at_horizon_le_eta_paper,
 baryogenesis_same_normalization_as_omega_k, ...)

DynamicBBNBaryogenesis (bbnBindingReleaseFactor, *_effectiveAtT,
 dynamicBBNShellReactionOpportunity,
 bbnShellReactionOpportunity_with_binding_feedback)
--> BBNEpochNetwork.bbnShellReactionOpportunity

GRFromMaxwell (same_phi / same_alpha / HQVM Friedmann compatibility)

BigBangNucleosynthesis (companion hqiv-bbn-paper; not in this note's graph)
--> BBNNetworkFromWeights, BBNEpochNetwork, BBNEpochEvolution, BBNWitness

Python witnesses: dynamic_bulk_bbn_v2.json, bbn_witnesses_dynamic.json

```

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